

RACHIT KUMAR TIWARI

CSE Undergraduate | Software & AI Developer

Kolkata, West Bengal | +91 9929243215 | tiwari.rachit@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

TECHNICAL SUMMARY

Computer science engineering student with hands-on experience in designing and developing AI-powered, full-stack web applications. Proficient in Java, Python, C, JavaScript, TypeScript, React, Next.js, Firebase, and REST APIs, with a strong focus on building scalable, user-centric software solutions. Experienced in developing end-to-end products, participating in hackathons, and creating AI-driven and social-impact platforms. Passionate about software engineering, problem-solving, and leveraging modern technologies to build innovative real-world applications.

TECHNICAL SKILLS

Programming Languages	Java, JavaScript, TypeScript, Python, C
Frontend	React, Next.js, HTML5, CSS3, Tailwind CSS, shadcn/ui
Backend	Node.js, REST APIs, WebSockets
Databases	MongoDB, Firebase, Supabase
AI & Computer Vision	OpenAI API, OCR, OpenCV, Face Recognition
Tools & Platforms	Git, GitHub, Postman, Figma, Vercel, Microsoft Word, Excel, PowerPoint
Core Concepts	OOP, DSA, API Design, Real-Time Systems, Responsive Web Design
Soft Skills	Problem-solving, Team Collaboration, communication, and Adaptability

PROJECT EXPERIENCE

Vocalis OS: An Agentic AI Operating System

Tech Stack: Next.js, TypeScript, React, Firebase, AI APIs

- Developed a voice-first AI operating system with speech recognition, text-to-speech, and autonomous task execution.
- Integrated AI-powered workflows, real-time web search, reminders, and to-do management.
- Designed a scalable, modular architecture supporting extensible AI agents and responsive user experiences.

OptraFi: An AI-Powered Personal Finance Platform

Tech Stack: React, TypeScript, Firebase, AI APIs

- Built an AI-powered personal finance platform for tracking income, expenses, and spending insights.
- Developed intelligent investment recommendation features using AI-driven financial analysis.
- Implemented secure, real-time dashboards with scalable cloud-based data management.

Vitamend: An AI-Driven Medicine Donation Platform

Tech Stack: React, TypeScript, MongoDB, OCR, AI Verification

- Built a medicine donation platform enabling secure redistribution of unused medicines through AI-assisted verification.
- Integrated OCR and AI pipelines for medicine identification and expiry validation.
- Designed modular donation, verification, and QR-based traceability workflows.

PrivSight: An AI Desktop Privacy System

Tech Stack: Electron, React, TypeScript, Python, OpenCV, WebSockets

- Engineered a desktop privacy application that protects sensitive on-screen content using real-time facial recognition.
- Built a computer vision pipeline with OpenCV and low-latency Electron–Python communication via WebSockets.
- Enabled fully local processing to ensure user privacy without cloud dependency.

Kred.ai: An AI-Powered Real-Time News Verification Platform

Tech Stack: Next.js 16, TypeScript, Google Gemini API, Serper API, Tailwind CSS

- Developed an AI-powered fact-checking platform that verifies news using live web evidence from trusted sources.
- Built a retrieval-augmented generation (RAG) pipeline integrating Serper API with Google Gemini for automated analysis.
- Implemented multilingual verification with evidence-backed verdicts, confidence scoring, and a responsive modern interface.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

MCKV Institute of Engineering, West Bengal, India

Year 1 (YGPA): 8.12/10

Semester 3 (SGPA): 8.80/10

Semester 2 (SGPA): 7.66/10

Semester 1 (SGPA): 8.64/10

CERTIFICATIONS

HackerRank – SQL (Advanced)

HackerRank – SQL (Intermediate)

HackerRank – SQL (Basic)

HackerRank – Java (Basic)

HackerRank – Python (Basic)

HackerRank – JavaScript (Intermediate)

HackerRank – JavaScript (Basic)

HackerRank – Node.js (Intermediate)

HackerRank – Node.js (Basic)

HackerRank – REST API (Intermediate)

HackerRank – Problem Solving (Intermediate)

HackerRank – Problem Solving (Basic)

HackerRank – CSS (Basic)